

Vitor Gama

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PROFESSIONAL SUMMARY

Chemical Engineer with extensive experience in process systems engineering, cost estimation, process simulation, optimization, and operability analysis. Skilled in coding and computational tools, bringing a strong technical edge to chemical engineering applications. A proactive and communicative team player, always open to discussing ideas, organizing events, and fostering a collaborative and engaging work environment. Strong leadership mindset with a focus on efficiency, problem-solving, and driving innovation through both technical expertise and team dynamics.

TECHNICAL PROFICIENCIES

- **Programming:** Python, MATLAB, C#, JavaScript, LaTeX.
- **Process Simulation:** AVEVA Process Simulation, Aspen Plus, Aspen Custom modeler, HYSYS, PRO/II, ChemCad.
- **Version Control:** Git, GitHub, GitLab.
- **Languages:** English, Portuguese.

EDUCATION

West Virginia University <i>Ph.D. candidate, Chemical Engineering</i> GPA: 3.75/4.00	Morgantown, WV Aug. 2021 – Dec. 2025 (Expected)
Federal University of Campina Grande <i>M.Sc., Chemical Engineering</i> GPA: 9.14/10.00	Campina Grande, Paraiba, Brazil Sept. 2018 – April 2021
Federal University of Campina Grande <i>B.Sc., Chemical Engineering</i> GPA: 8.18/10.00	Campina Grande, Paraiba, Brazil May 2013 – Aug. 2018

AWARDS & ACHIEVEMENTS

- AIChE Environmental Division Graduate Paper Award (2024) – 2nd place.
- Jack and Marietta Mullenger Fellowship (2024)
- Statler College PhD Recruitment Fellowship (2021)

RESEARCH EXPERIENCE

West Virginia University <i>Graduate Research Assistant (Ph.D.)</i>	Morgantown, WV Aug. 2021 – Present
• Design and simulation of gas separation membranes under Direct Air Capture (DAC) conditions. • Applied inverse design and process operability analysis to optimize system performance. • Published in peer-reviewed journals, demonstrating commitment to quality and research integrity.	
Federal University of Campina Grande <i>Graduate Research Assistant (M.Sc.)</i>	Campina Grande, Paraiba, Brazil Sept. 2018 – April 2021
• Designed and implemented CostApp, a cost estimation tool, which enhances cost analysis and engineering decision-making efficiency. • Developed process control routines in collaboration project with Petrobras, focusing on automation enhancements.	

RESEARCH PUBLICATIONS

- [1] **Vitor Gama et al.** “Connecting Material Characteristics with System Properties for Membrane-based Direct Air Capture (m-DAC) using Process Operability and Inverse Design Approaches”, **Industrial Engineering and Chemistry Research**. (under review).
- [2] **Vitor Gama et al.** “Process Operability Analysis of Membrane-Based Direct Air Capture for Low-Purity CO₂ Production”, **ACS Engineering Au**, 4.4 (Aug. 2024), pp. 394-404. DOI: 10.1021/acsengineeringau.3c00069.
- [3] **Vitor V. Gama et al.** “Modeling and Process Operability Analysis of a Direct Air Capture System”, **IFAC-PapersOnLine**, 13th IFAC Symposium on Dynamics and Control of Process Systems, including Biosystems DYCOPS 2022, Vol. 55.7 (Jan. 2022), pp. 316-321. DOI: 10.1016/j.ifacol.2022.07.463. ISSN: 2405-8963.

CONFERENCE PRESENTATIONS

- [1] **Vitor Gama**, Oishi Sanyal, Fernando V. Lima. *Leveraging Process Operability Mapping to Support Experimental Membrane Direct Air Capture (m-DAC) Solutions*. 2024 AIChE Annual Meeting, San Diego, CA, October 29, 2024 (Oral Presentation)
- [2] **Vitor Gama**, Oishi Sanyal, Fernando V. Lima. *Process Operability Analysis of Membrane-Based Direct Air Capture for Low-Purity CO₂ Production*. Membranes: Materials and Processes – Gordon Research Conference, New London, NH, July 28 – August 2, 2024 (Poster Presentation)
- [3] **Vitor Gama**, Oishi Sanyal, Fernando V. Lima. *Evaluation of Hollow Fiber Membrane Modules for CO₂ Direct Air Capture and Utilization in Low-Purity CO₂ Processes*. 2023 AIChE Annual Meeting, Orlando, FL, November 7, 2023 (Oral Presentation)
- [4] **Vitor Gama**, San Dinh, Oishi Sanyal, Fernando V. Lima. *Feasibility Analysis of a Membrane System for Direct Air Capture of CO₂*. 2022 AIChE Annual Meeting, Phoenix, AZ, November 17, 2022 (Oral Presentation)
- [5] **Vitor Gama**, Oishi Sanyal, Fernando V. Lima. *Designing a Membrane-based Platform for CO₂ Direct Air Capture Systems Analysis*. Gordon Research Conference on Chemical Separations, Ventura, CA, October 2–7, 2022 (Poster Presentation)

PROFESSIONAL LEADERSHIP & SERVICE

- Research Group Leader, CODES (2022 – Present).
 - Developing activities related to research group organization, lab visits and presentations.
- Research mentor, CODES (2022 – Present).
 - Mentored 3 undergraduate chemical engineering students, providing guidance in process simulation platforms, technical skills, and academic research development
- President, Brazilian Student Association (2022 – 2024).
 - Helped students connect with other brazilians in town and engaged with the community during events by presenting elements of brazilian culture for the general population.
- Member, Statler College DEI Student Committee (2022 – 2023).
 - Promoted events that fostered diversity on campus.

CLIFTONSTRENGTHS

Arranger | Positivity | Input | Woo | Communication